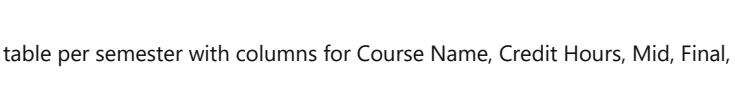
MCKRU CGPA Calculator

A modern, offline-first Flutter application for university students to calculate GPA and CGPA





Overview

MCKRU CGPA Calculator is a feature-rich mobile application built with Flutter that helps university students track and compute their academic performance. The app supports two calculation modes, a detailed marks sheet with auto-grade computation, and persists all data locally with zero network dependency.

Developer: Jibran Khan Dehpal
Package: `com.jibrankhan.cgpacalculator`
Version: 1.0.3+5

Key Features

Dual Calculation Modes

- **GPA (Single) Mode** — Enter individual courses with credit hours and marks. The app auto-converts marks to letter grades and computes semester GPA in real-time.
- **CGPA (Multi) Mode** — Enter multiple semesters with their GPAs and credit hours to compute cumulative CGPA.

Marks Sheet

- Full data-entry table per semester with columns for Course Name, Credit Hours, Mid, Final, Sessional, Total, and Grade.
- **Auto-calculation:** Mid + Final + Sessional = Total; Total auto-derives the Letter Grade.
- Tabbed semester navigation with add/delete capabilities.

Smart Marks-to-Grade Conversion

10-grade scale with support for practical/lab courses:

Grade	Points	Min Marks
A	4.0	85

A-	3.7	80
B+	3.3	75
B	3.0	70
B-	2.7	65
C+	2.3	61
C	2.0	58
C-	1.7	55
D	1.0	50
F	0.0	< 50

- **Lab/Practical toggle** — Doubles marks for grade calculation, simulating a 50-mark scale.

Real-time Stats Dashboard (Bento Grid)

- Color-coded CGPA display: Green (>= 3.5), Amber (>= 2.5), Red (< 2.5)
- Total credits tracking
- Updates instantly as data changes

Local-First Persistence

- All data saved via `SharedPreferences` as JSON
- Three independent storage stores for main data, CGPA results, and marks sheet
- Full offline support — zero network calls

Architecture & Tech Stack

Project Structure

```
lib/
├── main.dart                    # App entry point
├── models/
│   ├── course.dart              # Course data model
│   ├── semester.dart            # Semester model (holds courses)
│   └── semester_result.dart     # Semester result model (GPA + credits)
├── providers/
│   └── semester_provider.dart   # Riverpod state management
├── screens/
│   ├── home_screen.dart         # Main calculator screen
│   └── marks_sheet_screen.dart  # Detailed marks entry screen
├── services/
│   └── gpa_service.dart         # Grade/marks conversion logic
├── theme/
│   └── app_theme.dart           # Dark theme configuration
└── widgets/
    ├── bento_card.dart          # Stats summary card
    ├── course_row.dart          # Course input row
```

```
    ├── glass_card.dart          # Glassmorphism card
    ├── mesh_gradient_background.dart  # Gradient background
    ├── neomorphic_button.dart   # Styled button
    ├── semester_row.dart        # Semester input row
    └── semester_selector.dart   # Semester chip selector
```

State Management — Riverpod

Provider	Purpose
`semesterProvider`	Main GPA/CGPA state with courses and mode
`cgpaProvider`	Multi-semester CGPA results
`marksSheetProvider`	Independent marks sheet state
`gpaStatsProvider`	Derived real-time stats (computed)
`marksStatsProvider`	Derived marks sheet stats

Design Patterns

- **Immutable State** — All models use `copyWith` for state updates
- **Provider Pattern** — Clean separation of UI and business logic
- **JSON Serialization** — Full `toJson()` / `fromJson()` on all models
- **Component-Based UI** — Reusable widgets (`GlassCard` , `BentoCard` , etc.)

Dependencies

Package	Purpose
`flutter_riverpod`	State management
`google_fonts`	Lexend & Inter font families
`flutter_animate`	Declarative animations
`shared_preferences`	Local persistence
`uuid`	Unique ID generation

UI/UX Design

Theme

- **Material 3** dark theme with cyan primary and indigo secondary
- **Fonts:** Lexend (headings) + Inter (body) via Google Fonts
- **Background:** Mesh gradient with indigo and violet radial glows

Visual Effects

- **Glassmorphism** — Frosted glass cards with backdrop blur and subtle borders
- **Neomorphism** — Glowing indigo buttons with shadow effects
- **Mesh Gradient** — Multi-layered radial gradient background
- **Haptic Feedback** — Tactile responses on interactions (medium, light, heavy impact)

Color Coding

Color	Meaning
Emerald (Green)	GPA >= 3.5 — Excellent
Amber	GPA >= 2.5 — Good
Red	GPA < 2.5 — Needs Improvement

How It Works

GPA Mode Flow

1. Select "GPA (Single)" mode
2. Add courses with name, credit hours (1-6), and total marks (0-100)
3. Marks auto-convert to letter grades using the 10-grade scale
4. Semester GPA calculated as: `Σ(grade_points × credits) / Σ(credits)`
5. Overall CGPA computed across all semesters

CGPA Mode Flow

1. Select "CGPA (Multi)" mode
2. Add semesters with name, credit hours, and GPA (0.0-4.0)
3. CGPA computed as simple average of all semester GPAs

Marks Sheet Flow

1. Navigate to Marks Sheet from the home screen
2. Enter per-course marks: Mid, Final, Sessional
3. Total auto-sums; Grade auto-derives
4. Add/delete semesters and courses as needed

Platform Support

Platform	Status
Android	Signed release build configured
Windows	MSIX package configured
Web	Supported
iOS	Platform files present

Performance Optimizations

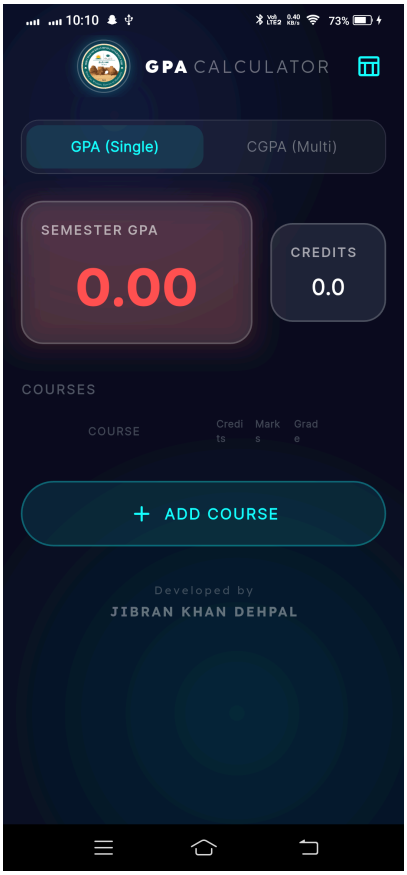
- `RepaintBoundary` on gradient background and stats grid
- Backdrop blur disabled on static cards
- Focus-node based input updates (not per-keystroke)
- Input values clamped at both widget and provider levels

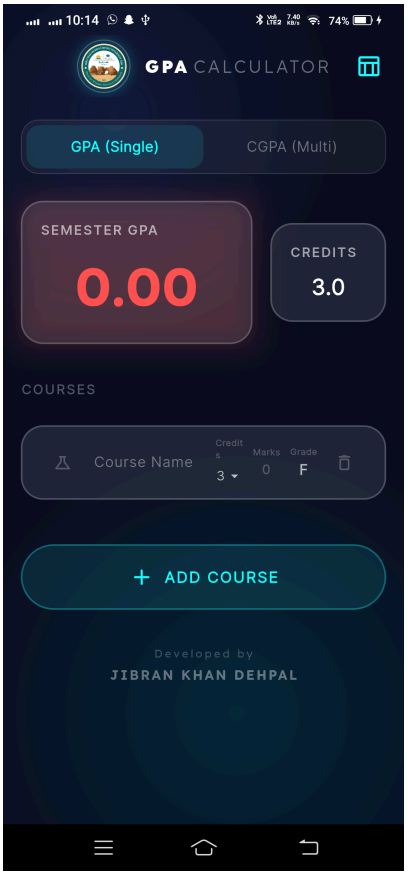
Privacy

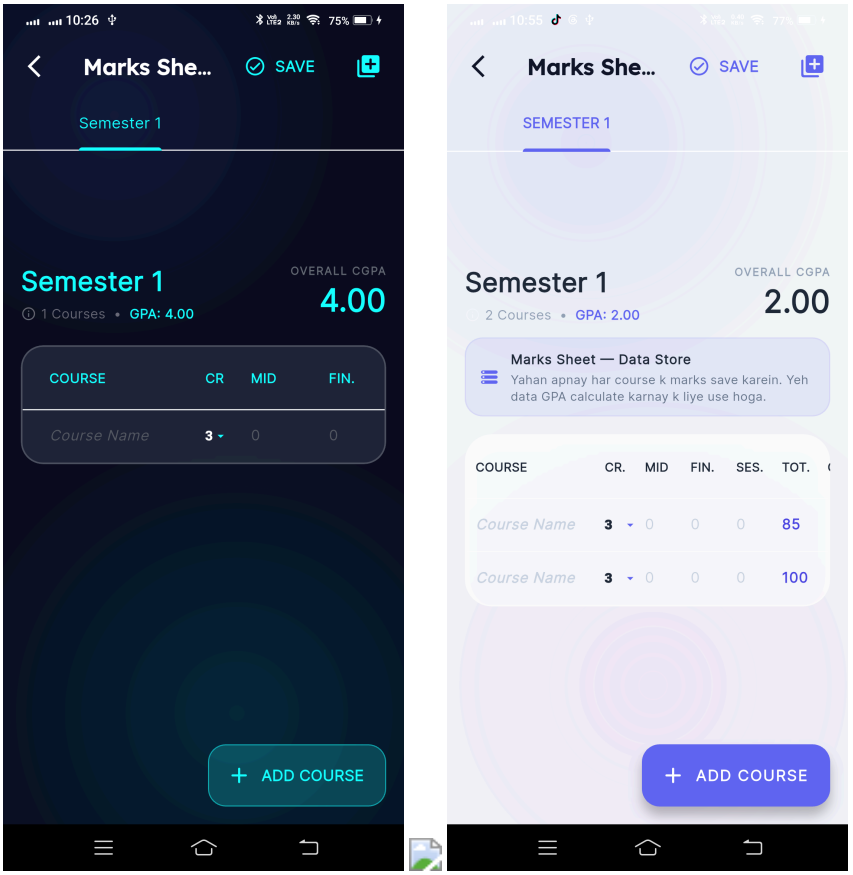
- **100% Offline** — No internet permission required
- **No Data Collection** — Zero analytics, zero tracking
- **Local Storage Only** — All data stays on device via SharedPreferences

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Screenshots


10:10
73%
GPA CALCULATOR
GPA (Single)
CGPA (Multi)
SEMESTER GPA
0.00
CREDITS
0.0
COURSES
COURSE
Credits
Marks
Grade
ADD COURSE
Developed by
JIBRAN KHAN DEHPAL


10:14
74%
GPA CALCULATOR
GPA (Single)
CGPA (Multi)
SEMESTER GPA
0.00
CREDITS
3.0
COURSES
Course Name
Credits
3
Marks
0
Grade
F
ADD COURSE
Developed by
JIBRAN KHAN DEHPAL



Getting Started

```
# Clone the repository
git clone https://github.com/your-username/cgpa_calculator.git

# Navigate to project directory
cd cgpa_calculator

# Install dependencies
flutter pub get

# Run the app
flutter run
```

Author

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> *Built with Flutter, Riverpod, and a passion for clean UI/UX design.*